

IDENTRA

AI-Powered Identity & Document Screening System

Complete Project Tech Stack Overview

Architecture Overview

- **Frontend**: Modern, fast, responsive Single Page Application (SPA)
- **Backend**: High-performance asynchronous REST API
- **AI / Computer Vision**: Embedded Python processing pipelines
- **Database / Auth**: Managed BaaS (Supabase)

Frontend Tech Stack

The User Interface and Experience layer

- **Core:** React 19 + TypeScript
- **Build Tool:** Vite (Fast HMR and Optimized builds)
- **Styling:** Tailwind CSS v4 (Utility-first CSS)
- **Routing:** React Router DOM
- **Animation:** Framer Motion
- **Icons:** Lucide React
- **Network:** Axios (API client)
- **State/Notifications:** React Hot Toast

Backend Tech Stack

The Server and API layer

- **Language:** Python 3
- **Framework:** FastAPI (High performance, async REST framework)
- **Server:** Uvicorn (ASGI server)
- **Data Validation:** Pydantic v2
- **Environment:** python-dotenv

AI & Processing Stack

The Document Analysis and Verification layer

- **Computer Vision:** OpenCV (`opencv-python-headless`)
- **Image Manipulation:** Pillow (PIL)
- **Numerical Computation:** NumPy
- **Cryptography:** `cryptography` , `PyJWT` (for secure data handling)

Database & Authentication

The Persistence layer

- **Provider:** Supabase
- **SDK:** `supabase-py` (Python SDK)
- **Features Used:**
 - Supabase Auth (Authentication)
 - Supabase Storage (Document Storage)
 - PostgreSQL (Underlying Database via PostgREST)

Development & Tooling

- **Code Formatting / Linting:** Oxlint (Frontend JS/TS)
- **Type Checking:** TypeScript (`tsc`)
- **Version Control:** Git
- **API Documentation:** Swagger UI / ReDoc (Auto-generated by FastAPI)

Summary

IDENTRA leverages a **React/Tailwind** frontend for a premium enterprise UI, paired with a blazing-fast **FastAPI** Python backend. Heavy lifting for document processing is handled natively by **OpenCV/NumPy**, while **Supabase** handles secure persistence and authentication.